CS 70 Discrete Mathematics and Probability Theory
Fall 2025 Sabin, Hug

HW 00

Due: Saturday, 8/30, 4:00 PM
Grace period until Monday, 9/01, 11:59 PM
Remember to show your work for all problems!

Sundry

Before you start writing your final homework submission, state briefly how you worked on it. Who else did you work with? List names and email addresses. (In case of homework party, you can just describe the group.) If you used an LLM, place transcripts of your chats here.

1 Administrivia

(a) Make sure you are on the course Ed (for Q&A) and Gradescope (for submitting homeworks, including this one). Find and familiarize yourself with the course website. What is its homepage's URL?

(b) Read the policies page on the course website.

  (i) What is the breakdown of how your grade is calculated?

  (ii) What is the attendance policy for discussions?

  (iii) When are homeworks released, and when are they due?

  (iv) How many "drops" do you get for homeworks? How many mini-vitamins will contribute to your grade?

  (v) When is the midterm? When is the final?

  (vi) What percentage score is needed to earn full credit on a homework?

2 Course Policies

Go to the course website and read the course policies carefully. Leave a followup on Ed if you have any questions. Are the following situations violations of course policy? Write "Yes" or "No", and a short explanation for each.

(a) Alice and Bob work on a problem in a study group. They write up a solution together and submit it, noting on their submissions that they wrote up their homework answers together.

CS 70 离散数学与概率论

2025 年秋季 萨宾，拥抱 HW 00

截止日期：8 月 30 日星期六下午 4:00 宽限期至 9 月 1 日星期一晚上 11:59 请务必写出所有题目的解题步骤！

杂项

在开始撰写最终作业之前，请简要说明你的作业完成过程。你还和哪些人合作完成作业？请列出他们的姓名和邮箱地址。 如果是小组作业，只需描述小组即可。）如果你使用了 LLM（可能是指某种在线聊天工具或软件），请在此处附上聊天记录。

1 行政琐事

（a）请确保您已登录课程 Ed（用于问答）和 Gradescope（用于提交作业，包括本次作业）。找到并熟悉课程网站。它的主页网址是什么？

（b）阅读课程网站上的政策页面。

（i）你的成绩是如何计算的？

（二）讨论课的出席政策是什么？

（三）作业何时发布，何时截止？

（四）作业能给你多少“分数”？多少“小维生素”能帮你提高成绩？

（五）期中考试是什么时候？期末考试是什么时候？

（六）作业需要达到多少百分比的分数才能获得满分？

2 课程政策

请访问课程网站并仔细阅读课程政策。如有任何疑问，请在 Ed 上留言。以下情况是否违反课程政策？请填写“是”或“否”，并简要解释每种情况。

（a）爱丽丝和鲍勃在一个学习小组中一起解决一个问题。他们共同写出解决方案并提交，并在提交的作业中注明他们是一起完成作业答案的。

(b) Carol goes to a homework party and listens to Dan describe his approach to a problem on the board, taking notes in the process. She writes up her homework submission from her notes, crediting Dan.

(c) Erin comes across a proof that is part of a homework problem while studying course material. She reads it and then, after she has understood it, writes her own solution using the same approach. She submits the homework with a citation to the website.

(d) Frank is having trouble with his homework and asks Grace for help. Grace lets Frank look at her written solution. Frank copies it onto his notebook and uses the copy as a reference to write and submit his homework, crediting Grace.

(e) Heidi has completed her homework using LaTeX. Her friend Irene has been working on a homework problem for hours, and asks Heidi for help. Heidi sends Irene her PDF solution, and Irene uses it to write her own solution with a citation to Heidi.

(f) Joe found homework solutions before they were officially released, and every time he got stuck, he looked at the solutions for a hint. He then cited the solutions as part of his submission.

(g) Kai is struggling with one of their homework problems, so they take a screenshot of the problem and ask ChatGPT to solve it for them. They adapt ChatGPT's response to their own solution, and they include a link to their ChatGPT conversation in the homework's Sundry section.

3 Use of Ed

Ed is incredibly useful for Q&A in such a large-scale class. We will use Ed for all important announcements. You should check it frequently. We also highly encourage you to use Ed to ask questions and answer questions from your fellow students.

(a) Read the Ed Etiquette section of the course policies and explain what is wrong with the following hypothetical student question: "Can someone explain the proof of Theorem XYZ to me?" (Assume Theorem XYZ is a complicated concept.)

(b) When are the weekly posts released? Are they required reading?

(c) If you have a question or concern not directly related to the course content, where should you direct it?

4 Academic Integrity

Please write or type out the following pledge in print, and sign it.

> I pledge to uphold the university's honor code: to act with honesty, integrity, and respect for others, including their work. By signing, I ensure that all written homework I submit will be in my own words, that I will acknowledge any collaboration or help received, and that I will neither give nor receive help on any examinations.

(b) 卡罗尔去参加一个家庭作业辅导会，听丹在黑板上讲解他解决问题的方法，并记下了笔记。她根据笔记完成了自己的作业，并在作业中注明了丹的解题思路。

（c）艾琳在学习课程材料时，偶然发现了一道家庭作业题的证明。
她阅读了原文，理解之后，用同样的方法写出了自己的解答。她提交作业时，附上了网站的出处。

(d) 弗兰克做作业遇到困难，便向格蕾丝求助。格蕾丝让弗兰克看了她写好的答案。弗兰克把答案抄到笔记本上，并以此为参考完成作业，提交时注明了格蕾丝的贡献。

(e) 海蒂用 LTEX 完成了作业。她的朋友艾琳做了好几个小时的作业题，于是向海蒂求助。海蒂把她的 PDF 解答发给了艾琳，艾琳根据海蒂的解答写出了自己的答案，并注明了出处。

(f) 乔在作业答案正式发布前就找到了答案，每次遇到难题，他都会参考答案寻找线索。然后，他在提交的作业中引用了这些答案。

(g) Kai 在做作业时遇到一道难题，于是他截屏了题目，并请 ChatGPT 帮忙解答。他根据 ChatGPT 的解答修改了自己的答案，并将 ChatGPT 的对话链接添加到了作业的“其他”部分。

3 埃德的使用

在这样的大型课堂上，Ed 对于问答环节非常有用。我们将使用 Ed 发布所有重要通知，请大家经常查看。我们也强烈建议大家使用 Ed 向同学提问和回答问题。

（a）阅读课程政策中的“教育礼仪”部分，并解释以下假设的学生问题有何不妥之处：“有人能给我解释一下定理 XYZ 的证明吗？”（假设定理 XYZ 是一个复杂的概念。）

（b）每周的文章何时发布？是必读的吗？

（c）如果您有与课程内容没有直接关系的问题或疑虑，您应该向谁咨询？

4 学术诚信

请将以下承诺以书面或打字形式写下来并签名。

> 我承诺遵守大学的荣誉准则：诚实守信，尊重他人，包括他人的作品。我在此签字，保证我提交的所有书面作业均由我本人撰写，我会注明所有合作或获得的帮助，并且不会在任何考试中给予或接受帮助。

5 Propositional Practice

Note 1

In parts (a)–(b), convert the English sentences into propositional logic. In parts (c)–(d), convert the propositions into English. For parts (b) and (d), use the notation $a \mid b$ to denote the statement "a divides b", and use the notation $P(x)$ to denote the statement "x is a prime number".

(a) For every real number k, there is a unique real solution to $x^3 = k$.

(b) If p is a prime number, then for any two natural numbers a and b, if p doesn't divide a and p divides ab, then p divides b.

(c) $(\forall x, y \in \mathbb{R})[(xy = 0) \implies ((x = 0) \vee (y = 0))]$

(d) $\neg((\exists y \in \mathbb{N})[(\forall x \in \mathbb{N})[(x > y) \implies ((y \mid x) \vee P(x))]])$

5 命题实践

注 1 在 (a) 至 (b) 部分，将英文句子转换为命题逻辑。在 (c) 至 (d) 部分，将命题转换为英文。对于 (b) 和 (d) 部分，使用符号 $a \mid b$ 表示命题“a 整除 b”，使用符号 $P(x)$ 表示命题“x 是素数”。

（a）对于每个实数 k，方程 x=k 都有唯一的实数解。

（b）如果 p 是质数，那么对于任意两个自然数 a 和 b，如果 p 不能整除 a 且 p 能整除 ab，那么 p 能整除 b。

(c) $(\forall x, y \in R)[(xy = 0) \Longrightarrow ((x = 0) \vee (y = 0))]$

(d) $\neg((\exists y \in N)[(\forall x \in N)[(x > y) \Longrightarrow ((y \mid x) \vee P(x))]])$